Statistics & Probability

Permutations

1 How many ways can 5 distinct books be arranged on a shelf?

2 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

3 How many distinct arrangements are there of the letters in the word APPLE?

Combinations

4 How many ways can a committee of 4 be chosen from 11 people?

5 A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

6 From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

Probability rules

7 A fair die is rolled. Find $P(\text{even or greater than 4})$.

8 $P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

9 A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

Conditional probability and independence

10 $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

11 In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

12 Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

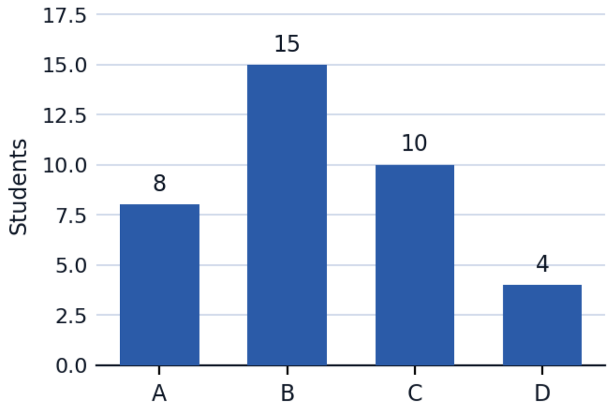
Measures of center and spread

13 Find the mean and median of the data set $12, 15, 18, 20, 25$.

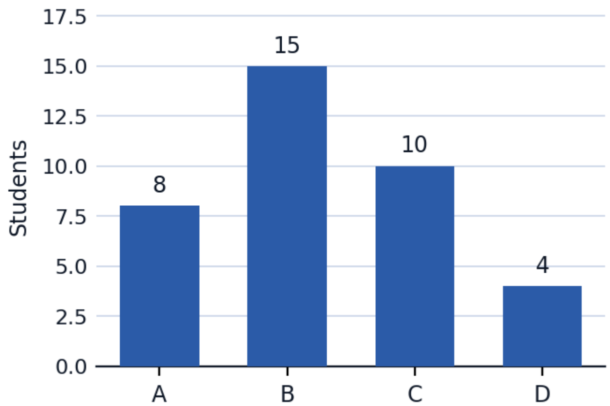
14 Find the population standard deviation of the data set $4, 8, 6, 5, 3$.

Data representation

15 The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.



16 Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.



The binomial theorem

17 Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.

18 In the expansion of $(2x - 1)^4$, find the coefficient of x^2.

The normal distribution

19 A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

20 For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

A well-designed word problem: quality control sampling

21 This question has three parts. A factory produces batteries, and 4% of batteries are defective. A quality-control inspector randomly selects a sample of 3 batteries from a large shipment (assume the selections are independent). (a) Find the probability that none of the 3 batteries are defective. (b) Find the probability that exactly 1 of the 3 batteries is defective. (c) Find the probability that at least 1 of the 3 batteries is defective.